

Task 2

Embed it

Embedding Operation is a technique used to convert words, items, or categories into meaningful numeric vectors and is widely applied in areas like NLP, recommendation systems, and search engines.

In this section, you will optimize the embedding operation using optimization techniques such as software prefetching and SIMD.

List of Tasks (8 points)		
2A	Software prefetching	5 points
2B	SIMD	1 points
2C	Software prefetching + SIMD	2 points

Task 2A: Software prefetching

In this part of the assignment, you will be optimizing the embedding operation code using the software prefetching technique. You will be tuning various parameters of software prefetching for the embedding operation by analyzing trends in different CPU performance metrics. Additionally, you will study the impact of hardware prefetchers on the performance of the embedding operation.



You will be tuning two key parameters of software prefetching:

1. Prefetch Distance
2. Cache Fill Level

Both **`_builtin_prefetch`** and **`_mm_prefetch`** allow you to control these parameters during software prefetching.

Additionally, you will use the **`sw_prefetch_access`** event in *perf* to track the number of software prefetch requests issued during the execution.

Deliverables

1. *Analyze the Impact of Embedding Table Size:*

- Increase the size of the embedding table and observe how software prefetching performance changes.
- Optionally, you can also experiment with varying the sizes of the input array and offset array.

2. *Analyze CPU Metrics:*

For different embedding table sizes and different software prefetching parameters, analyze the trends in key CPU metrics (use the *perf* tool):

- L1D Cache Misses
- L2 Cache Misses
- LLC (Last Level Cache) Misses

3. *Collect Execution Time and Compute Speedup:*

- Measure the execution time of the embedding operation with and without software prefetching.
- Compute the speedup as follows:

$$Speedup = \frac{Execution\ Time\ without\ Software\ Prefetching}{Execution\ Time\ with\ Software\ Prefetching}$$

- Analyze how the speedup varies with different embedding table sizes and different software prefetching parameters.

4. *Identify Optimal Parameters:*

Determine the optimal prefetch distance and optimal cache fill level based on your observations.

5. *Study the Effect of Hardware Prefetchers:*

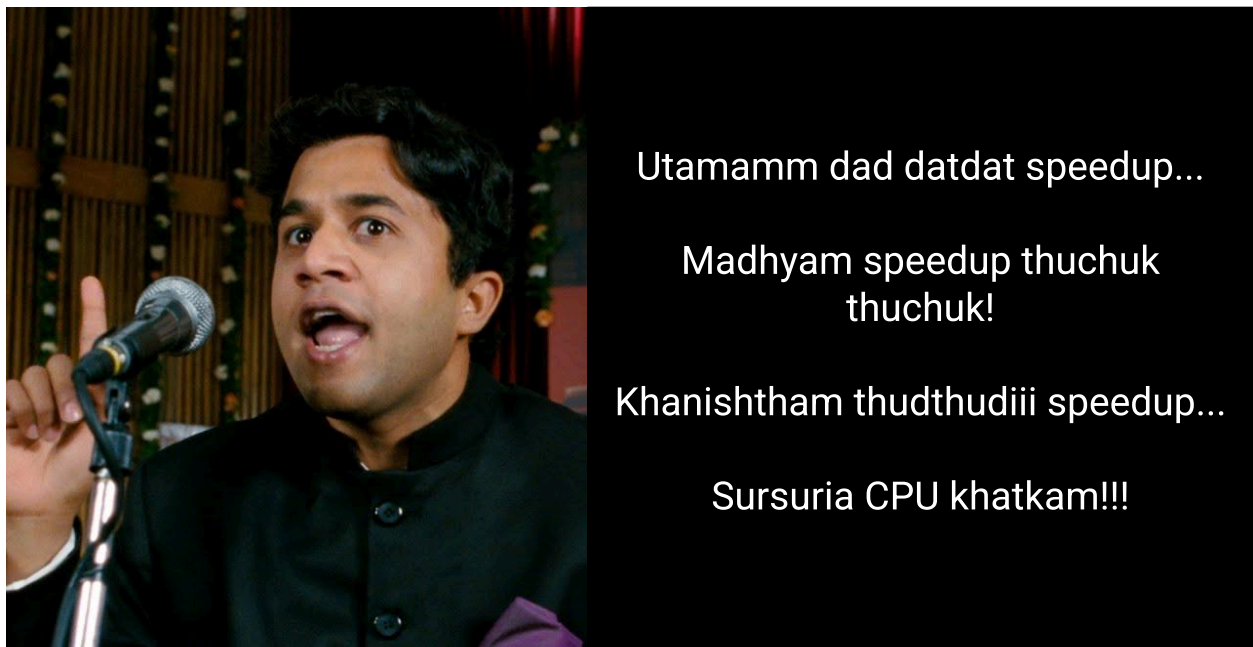
Enable and disable hardware prefetchers and analyze their impact on the embedding operation's performance.

Consider **Table 2.1** for reference. Identify the values of different metrics for different embedding table sizes, prefetching distances, and cache fill levels. Analyze them carefully and state your insights for the question below.

1. What trend do you observe in speedup with different embedding table sizes?

Speedup decreases slightly as embedding size grows.

2. What is the best prefetch distance?
4 gives more stable speedups across all sizes.
3. At what cache fill level do you achieve the maximum speedup
L1



Approach 2:

Running the prefetch loop for full embedding dimension.

Software_Prefetch Off:

Embedding Size	Cycles	Instructions	L1_Misses	L2_Misses	LLC_Misses	Execution Time
1000000	3697940918	12105765151	8086529.6	10668198	11095.6	133.8
2000000	7401673711	24209403551	16130490.6	21265047	13106.4	137.2
3000000	11102835826	36316175115	24169643.2	31944808.8	14669.6	144.6

Software_Prefetch ON:

Embedding Size	Prefetch Distance	Cache Level	Cycles	Instructions	L1 Misses	L2 Misses	LLC Misses	Execution Time	SW Prefetch Requests
1000000	4	L1	4210045332	12137779345	8099294.6	10680734.6	5427.6	78.2	5864.6
2000000	4	L1	8418352711	24273422853	16162670.6	21354575.6	7467.8	83.6	5864
3000000	4	L1	12630599144	36412188820	24218800.2	31978885.4	9920.4	86.8	5865
1000000	16	L1	4209937197	12137776461	8098756.2	10686555	7147	82.2	3944.8
2000000	16	L1	8421424538	24273422489	16158062.8	21318958.6	9728.6	85	3945.2
3000000	16	L1	12611873254	36412190950	24228283.2	31987134.6	15936.6	103.6	3945

1000000	4	L2	4208107419	12137790568	810431 8.8	10714772	5404.4	89.8	744
2000000	4	L2	8421368830	24273436160	161557 75	21299721. 2	7862.4	103.6	744
3000000	4	L2	1263200505 9	36412178375	242102 99.6	31915736. 2	10884	100.8	744
1000000	16	L2	4210166140	12137784557	809692 1.6	10688506	7261.6	82.8	744
2000000	16	L2	8433182255	24273430066	161786 20	21314565	9693.6	88.2	744
3000000	16	L2	1262260358 2	36412183944	242222 71.2	32014428. 4	13668.2	97.2	744
1000000	4	LLC	4208383912	12137778749	810031 0.4	10714255. 8	5275.2	87.2	744
2000000	4	LLC	8418035373	24273426844	161726 38.6	21360115. 4	7669	97	744
3000000	4	LLC	1263752440 5	36412198464	242317 93.6	31989290	14634	104.8	744
1000000	16	LLC	4208261341	12137780649	813927 4.4	10728507. 8	7112.4	82.2	744
2000000	16	LLC	8417869977	24273422226	161689 35	21368243. 6	9637.2	88.8	744
3000000	16	LLC	1262097336 2	36412176853	242442 60	31944651	12350.2	105.8	744

Embedding Size	Prefetch Distance	Cache Level	Speedup
1M	4	L1	1.71×
2M	4	L1	1.64×
3M	4	L1	1.67×
1M	16	L1	1.63×
2M	16	L1	1.61×
3M	16	L1	1.39×
1M	4	L2	1.49×
2M	4	L2	1.32×

3M	4	L2	1.43×
1M	16	L2	1.62×
2M	16	L2	1.55×
3M	16	L2	1.49×
1M	4	LLC	1.53×
2M	4	LLC	1.41×
3M	4	LLC	1.38×
1M	16	LLC	1.63×
2M	16	LLC	1.55×
3M	16	LLC	1.37×

Observations for Questions

Q1. Trend with different embedding table sizes:

Speedup is highest for smaller embedding tables (1M).

As embedding size grows (2M, 3M), speedup decreases slightly, likely because cache reuse reduces and prefetches cannot keep up with memory latency.

Q2. Best Prefetch Distance:

Distance = **4 gives more stable speedups across all sizes.**

Distance = 16 sometimes helps (1M, 2M) but degrades at larger embedding size (3M).

Q3. Optimal Cache Fill Level:

L1 consistently achieves the best or near-best speedup (~ 1.6 – $1.7\times$).

L2/LLC speedups are lower (~ 1.3 – $1.5\times$).

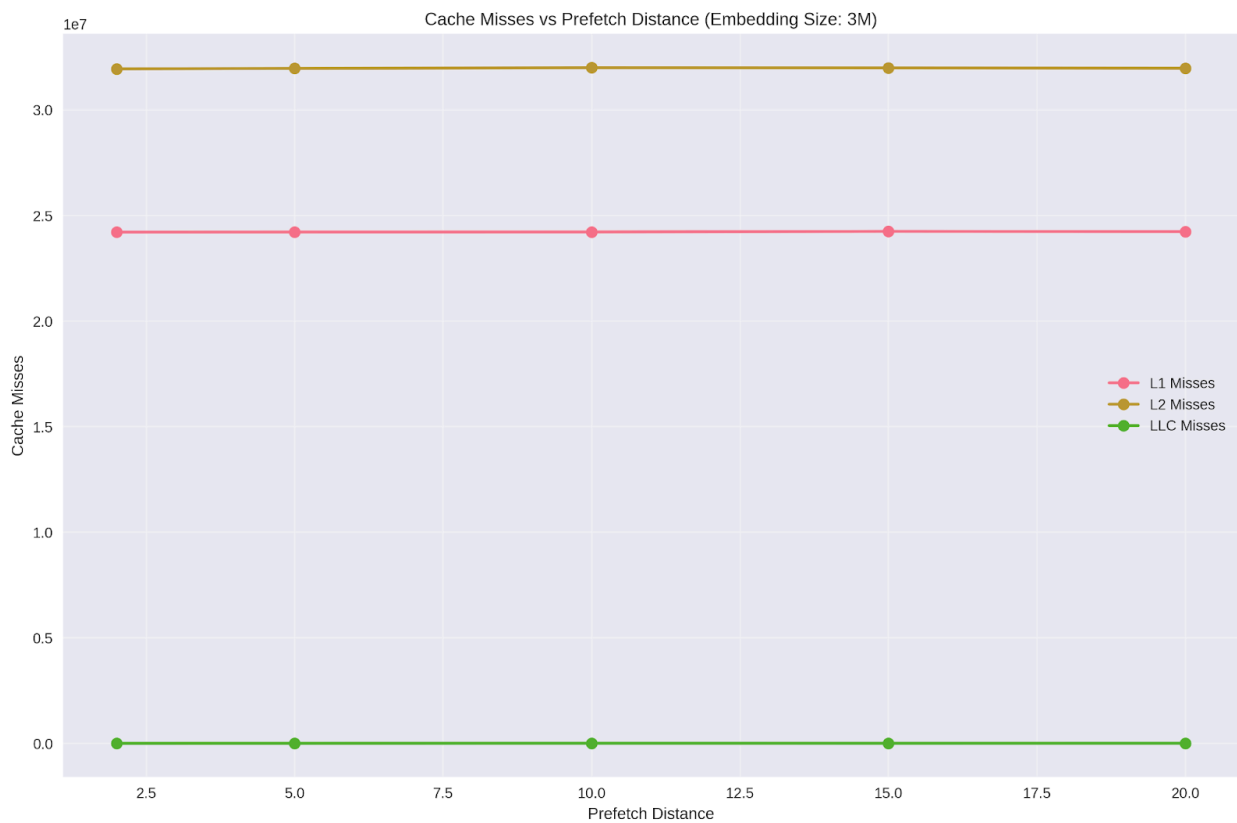
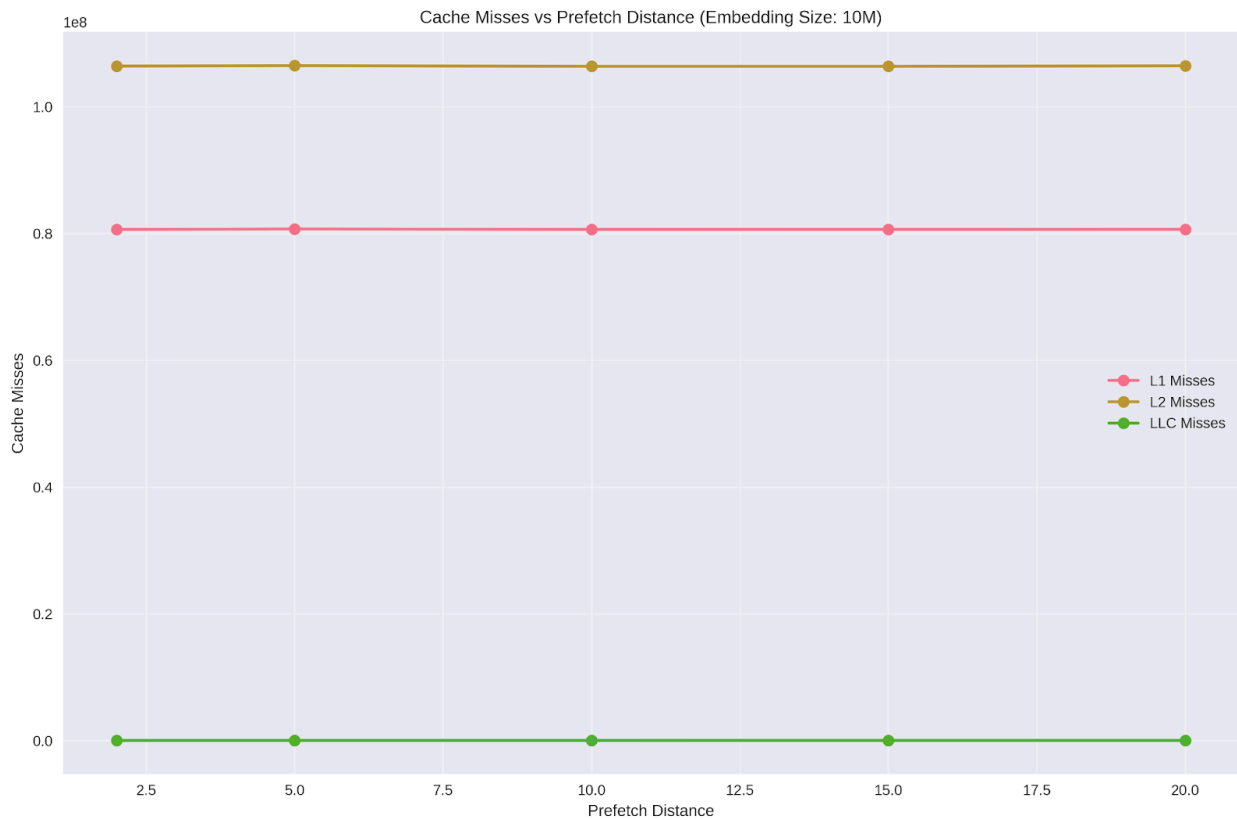
Thus, prefetching into L1 is optimal.

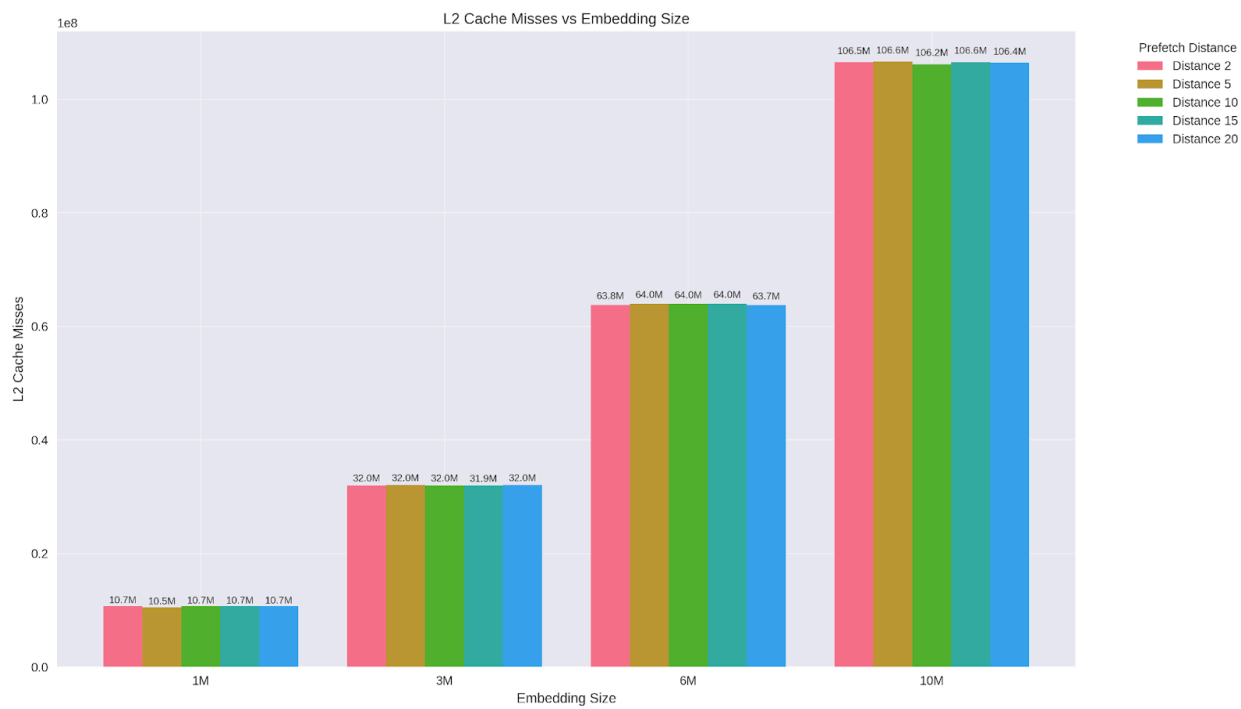
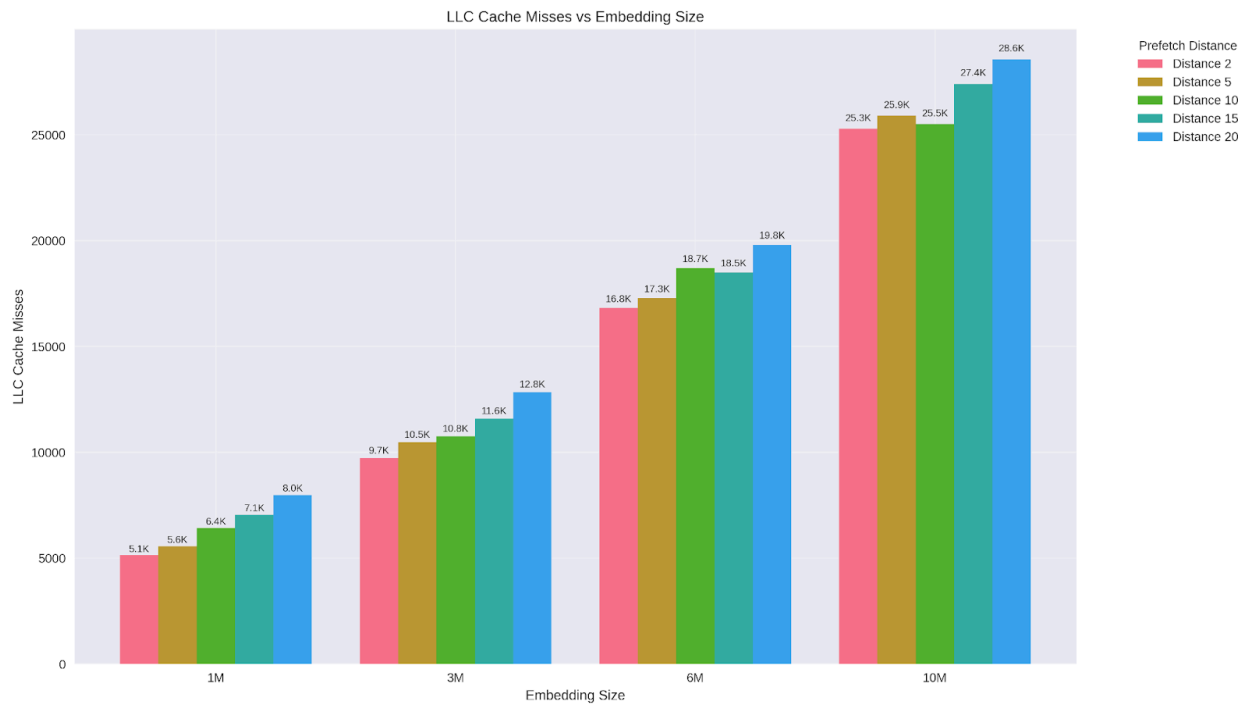
The below graph were generated when running the experiment for the below parameters

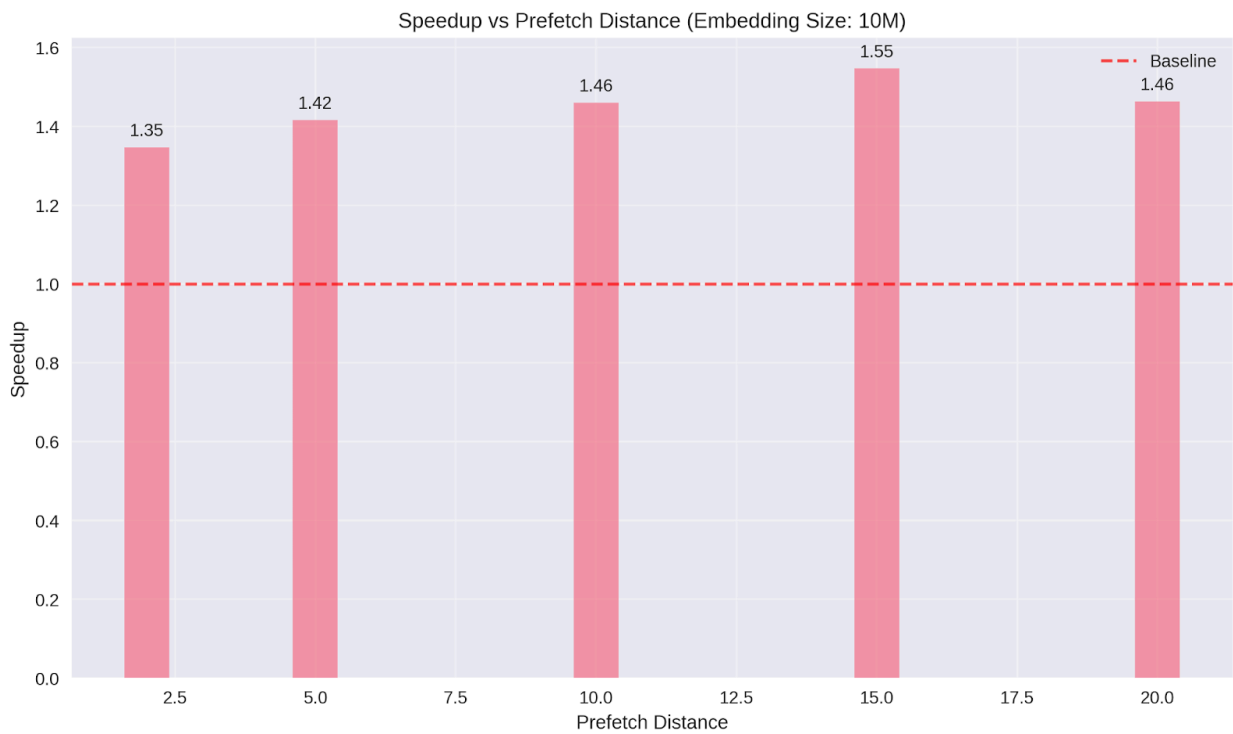
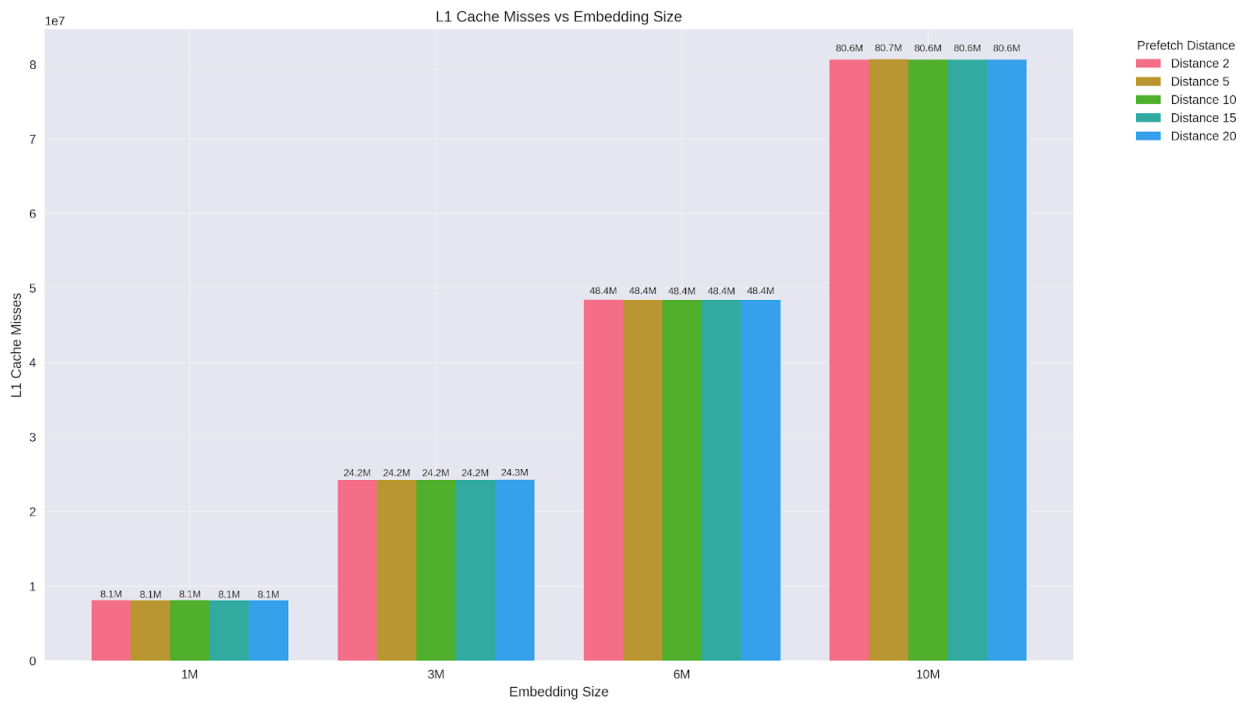
EMBED_SIZES=(1000000 3000000 6000000 10000000)

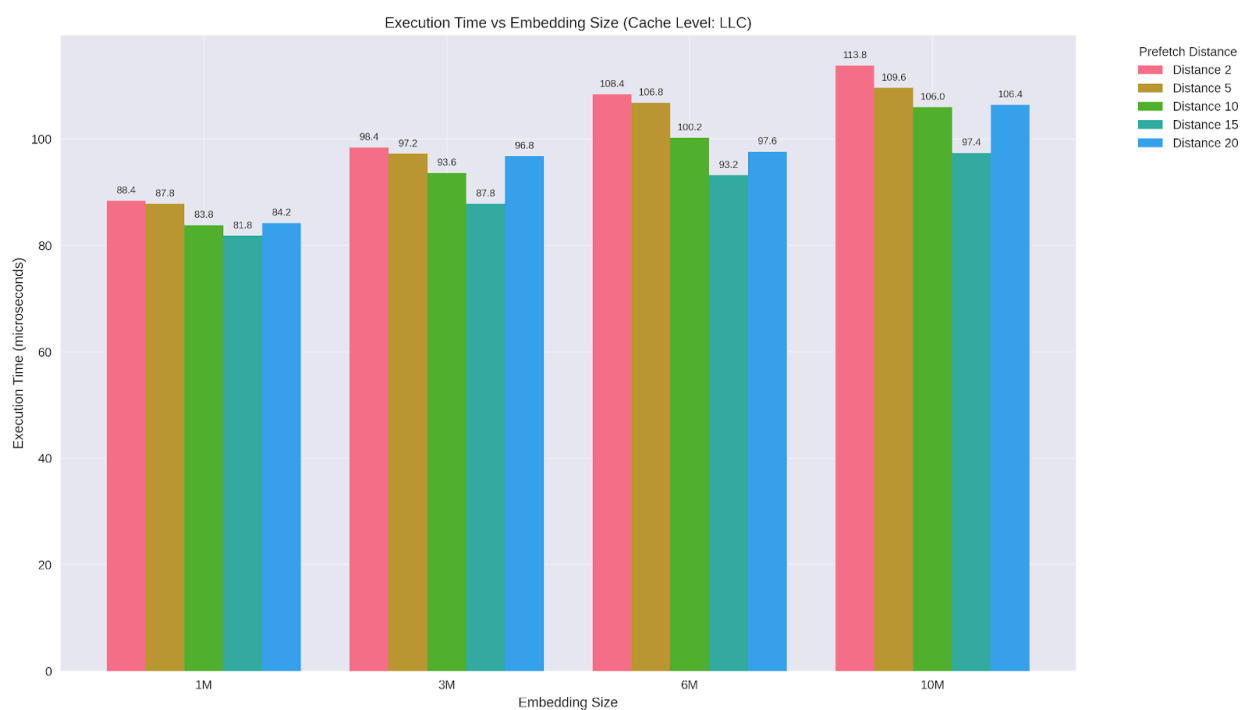
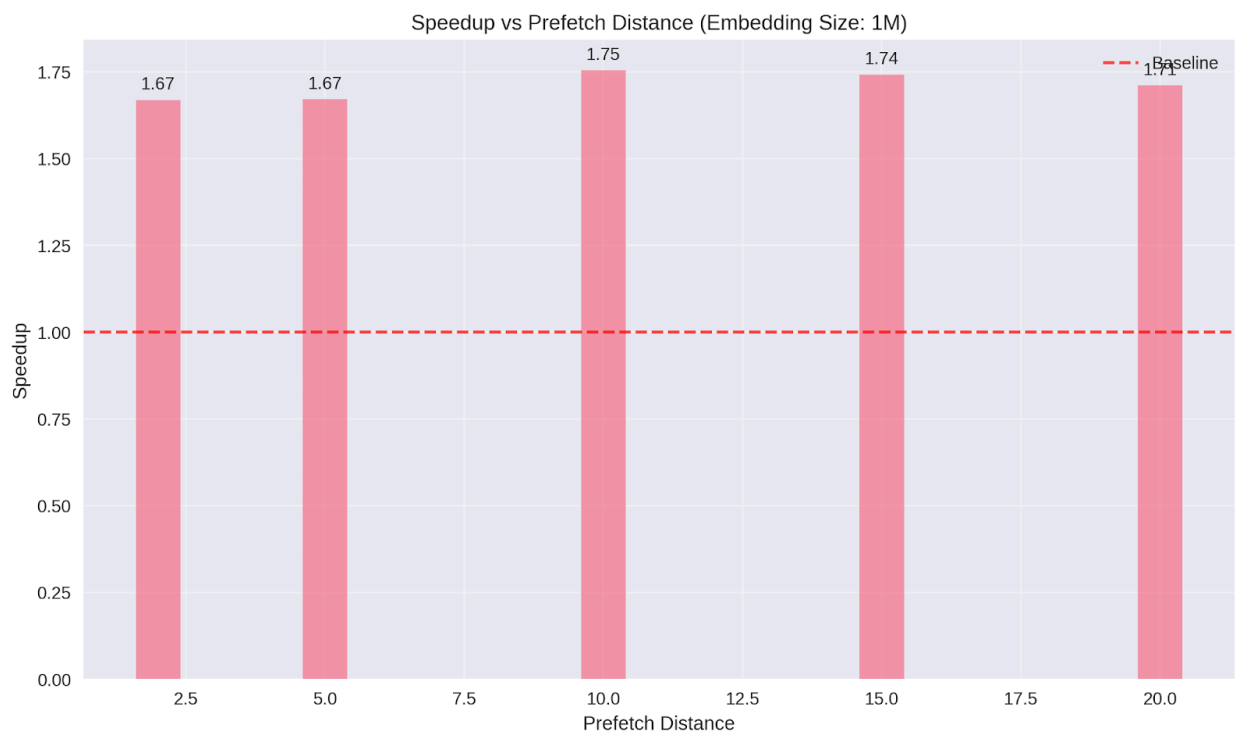
PREFETCH_DISTANCES=(2 5 10 15 20)

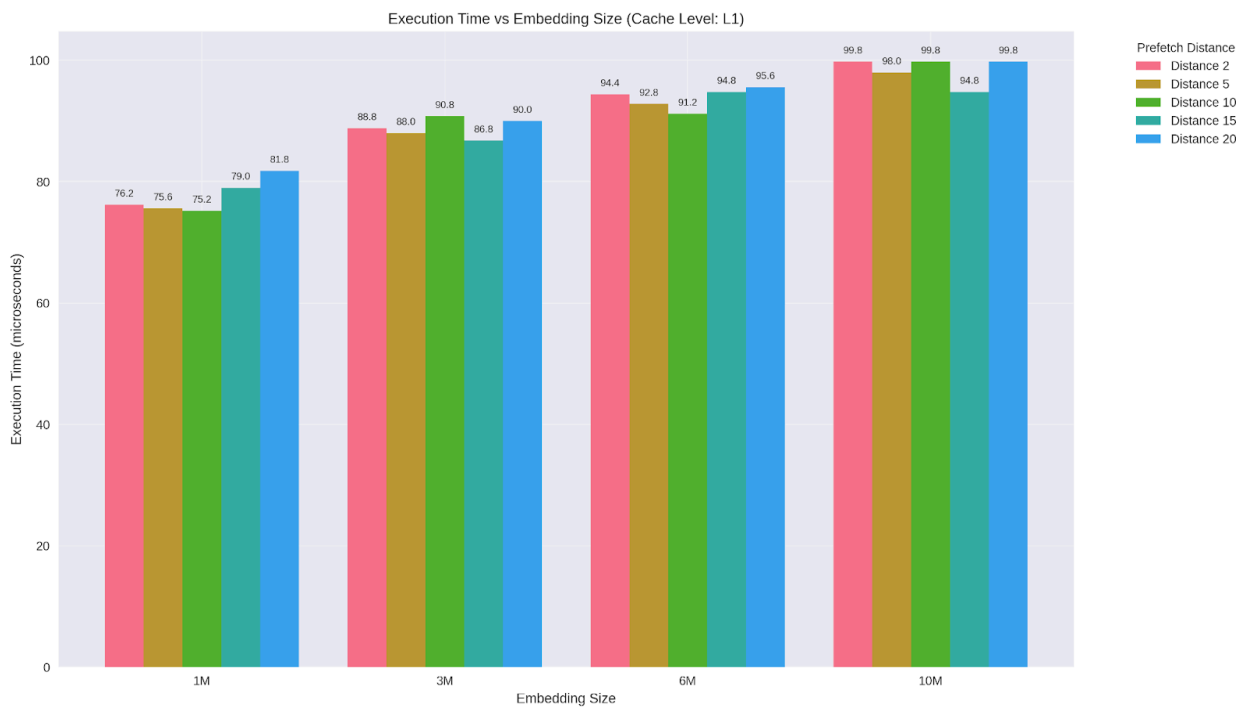
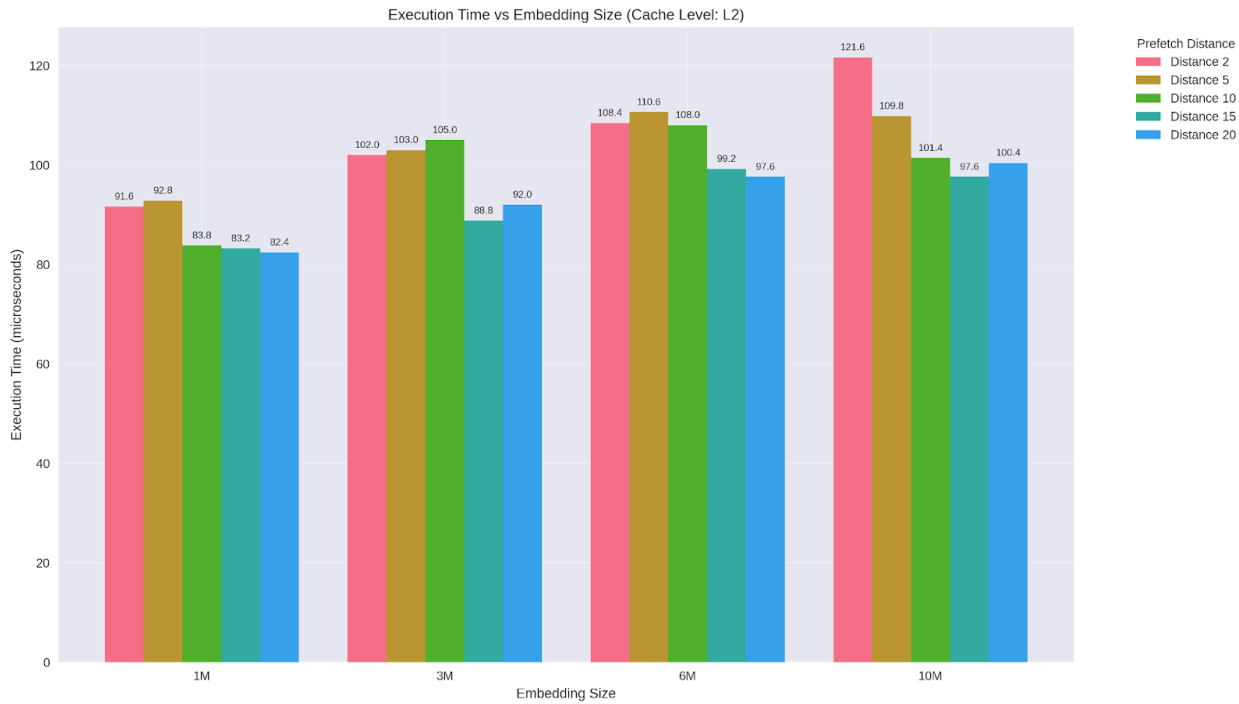
CACHE_LEVELS=("L1" "L2" "LLC")











Trend with size: Speedup decreases slightly as embedding size grows.

Approach 1: when the full embedding dimension was not fetched

Prefetching Mode	Embedding Table Size	Prefetch Distance	Cache Fill Level	L1D Misses	L2 Misses	LLC Misses	Software Prefetch Requests	Execution Time	Speedup
No Prefetching	1M	-	-	8159346	10861610	9738	0	418	1
Software Prefetching	1M	2	L1	8141840	10848635	9162	1465	342	1.222222222
		2	L2	8135076	10826921	7710	722	344	1.215116279
		2	LLC	8128386	10821661	8076	0	360	1.161756531
		4	L1	8145233	10846908	8511	1466	338	1.236686391
		4	L2	8134190	10829094	8997	722	346	1.20739457
		4	LLC	8177045	10880604	8425	0	347	1.203917051
		8	L1	8136770	10838276	8934	1464	345	1.210892236
		8	L2	8148252	10850365	5743	718	351	1.192241871
		8	LLC	8150670	10852209	5768	0	343	1.218658892
		16	L1	8134365	10822808	7385	1463	340	1.229411765
		16	L2	8138700	10838306	8544	719	362	1.154058531
		16	LLC	8182857	10858700	6577	0	338	1.235955056
		32	L1	8142906	10846605	8574	1463	340	1.228689006
		32	L2	8135018	10830980	7789	720	351	1.190883191
		32	LLC	8144918	10846145	6153	0	367	1.137724551

Prefetching Mode	Embedding Table Size	Prefetch Distance	Cache Fill Level	L1D Misses	L2 Misses	LLC Misses	Software Prefetch Requests	Execution Time	Speedup
No Prefetching	2M	-	-	24207108	32155296	10497	0	435	
Software Prefetching	2M	2	L1	16224527	21568555	8400	1467	357	1.219854178
		2	L2	16252679.4	21628812.6	9892.4	721.4	349	1.247133028
		2	LLC	16246996.6	21597453.2	9090.8	0	360	1.209677419
		4	L1	16243944.2	21599506.8	8982.2	1465.8	349	1.244991414
		4	L2	16261215.4	21582074.4	12037.6	722	349	1.244991414
		4	LLC	16232239	21574040.6	10627.8	0	356	1.221224031
		8	L1	16237460.2	21602848	8815.2	1463	352	1.234392736
		8	L2	16283089.2	21691222	9110.4	718.6	353	1.232294618
		8	LLC	16251071.6	21598987.6	8753.6	0	361	1.206322795
		16	L1	16215099.4	21564091	8921	1462	350	1.242857143
		16	L2	16237063.4	21596176	8925.4	719	350	1.243567753
		16	LLC	16224271.2	21559120.8	8848.2	0	359	1.212374582
		32	L1	16226054	21584366	8642	1463	353	1.232294618
		32	L2	16223045.4	21567675.4	9256.2	718.4	364.4	1.193743139
		32	LLC	16226449.6	21586093.6	9342.8	0	360	1.208333333

Prefetching Mode	Embedding Table Size	Prefetch Distance	Cache Fill Level	L1D Misses	L2 Misses	LLC Misses	Software Prefetch Requests	Execution Time	Speedup
No Prefetching	3M	-	-	24359791	32322661	9940	0	426	
Software	3M	2	L1	24311257	32324035	11857	1465	357	1.194615816

Prefetching									
		2	L2	24387535	32463714	1204 3	722	367	1.160762943
		2	LLC	24325301	32339919	11433	0	365	1.166484118
		4	L1	24373498	32353911	1369 0	1465	358	1.189944134
		4	L2	24291335	32293174	11447	721	370	1.150107991
		4	LLC	24320010	32330689	11346	0	356	1.196629213
		8	L1	24358445	32405625	1088 0	1463	353	1.206115515
		8	L2	24333718	32355292	11397	718	364	1.171617162
		8	LLC	24316519	32325554	11745	0	364	1.169045005
		16	L1	24372961	32412538	11416	1463	360	1.183333333
		16	L2	24398734	32340621	1474 5	718	363	1.174200662
		16	LLC	24382178	32352030	1506 1	0	358	1.191275168
		32	L1	24321541	32322865	1096 3	1463	357	1.193946188
		32	L2	24299282	32302163	1347 6	719	368	1.156351792
		32	LLC	24472446	32511089	1202 9	0	366	1.163934426

Prefetching Mode	Embedding Table Size	Prefetch Distance	Cache Fill Level	L1D Misses	L2 Misses	LLC Misses	Software Prefetch Requests	Execution Time	Speedup
No Prefetching	4M	-	-	32326345	42986923	69146	0	439	
Software Prefetching	4M	2	L1	32468743	43164193	13643	1465	358	1.224888393
		2	L2	32423785	43061859	14735	721	374	1.175053533
		2	LLC	32486117	43238125	14040	0	360.2	1.218767351

					4	6			
		4	L1	32439714.6	43102224	14454.8	1465.2	356.6	1.231071228
		4	L2	32488709	43240528.8	14262.6	721.2	369	1.189057421
		4	LLC	32433480.4	43089581.8	14310	0	355	1.238014664
		8	L1	32409089	43052577.8	14113.6	1463.6	357	1.229003359
		8	L2	32423588.4	43114811.4	13718.2	717.6	356	1.233839236
		8	LLC	32443971.6	43122030.4	17151.6	0	363	1.208035223
		16	L1	32410850.4	43073614.8	19099.4	1463	373	1.1769437
		16	L2	32424319.2	43088570.4	13990.2	718.8	356	1.233146067
		16	LLC	32450405.6	43167960.2	17009.6	0	364	1.206043956
		32	L1	32399122	43064834.2	19308.8	1462.6	365	1.201423098
		32	L2	32404336.4	43071333.4	14284.4	718.2	360	1.218767351
		32	LLC	32414132.4	43069847.6	14534	0	362	1.212037548

Prefetching Mode	Embedding Table Size	Prefetch Distance	Cache Fill Level	L1D Misses	L2 Misses	LLC Misses	Software Prefetch Requests	Execution Time	Speedup
No Prefetching	5M	-	-	40553288	53644038	16448	0	478	
Software Prefetching	5M	2	L1	40525062	53827006	17027	1464.6	365.2	1.308871851
		2	L2	40723233.2	54048963.6	16754.4	721	367.6	1.300326442
		2	LLC	40679854.2	54156323.	22340	0	373.6	1.279443255

					8				
		4	L1	40507484.8	53835325.6	17952.8	1465.4	360.6	1.325568497
		4	L2	40560232.4	53872109.8	18766.4	721.2	369.4	1.293990254
		4	LLC	40548880.8	53929042.4	21927.8	0	369.6	1.293290043
		8	L1	40507560.8	53834682.2	18465.6	1462.8	366	1.306010929
		8	L2	40639577.4	53907645.4	21304.6	718.2	367.6	1.300326442
		8	LLC	40487472.2	53815559.8	27219.4	0	368.6	1.296798698
		16	L1	40582467.2	53956070.8	18439.8	1462	369.6	1.293290043
		16	L2	40512166.6	53828511.4	16591.8	718	361	1.324099723
		16	LLC	40486339.6	53806480.4	17643.8	0	376	1.271276596
		32	L1	40580801.4	53856103.6	20224.8	1463.8	364	1.313186813
		32	L2	40437120.8	53750045.4	18529.6	720.2	368.2	1.298207496
		32	LLC	40520494.2	53863494	20368.8	0	371.2	1.287715517

Prefetching Mode	Embedding Table Size	Prefetch Distance	Cache Fill Level	L1D Misses	L2 Misses	LLC Misses	Software Prefetch Requests	Execution Time	Speedup
No Prefetching	6M	-	-	48330556	64222621	17950	0	443	
Software Prefetching	6M	2	L1	48590406	64575113.8	27014.4	1464.6	368.6	1.079006772
		2	L2	48600005	64635078.6	27662	720.8	362.6	1.296798698
		2	LLC	48573456	64585864.8	23441.8	0	375.6	1.318257033
		4	L1	48593729.8	64621253.8	20684.6	1465.2	364.4	1.272630458
		4	L2	48613018.4	64670204.	26065.	721	384.2	1.311745335

					6	4			
		4	LLC	48617667.6	64595812.4	22677.6	0	364.6	1.244143675
		8	L1	48756144.6	64698339	24076.6	1462.6	366.2	1.311025782
		8	L2	48653709.6	64692621.2	23473	720	363.2	1.305297652
		8	LLC	48620984.2	64604955	20539.2	0	362	1.316079295
		16	L1	48690646.8	64728045.8	19359	1462.6	368.8	1.320441989
		16	L2	48688071	64592111	27281.4	718.2	377.4	1.296095445
		16	LLC	48649566.4	64671835.2	27815	0	364	1.266560678
		32	L1	48627727.6	64677652.2	19736.4	1462.6	374	1.311745335
		32	L2	48588365.8	64607140.4	21196	719.4	367	1.279443255
		32	LLC	48573638.4	64580336.6	21079	0	368	1.302452316

Prefetching Mode	Embedding Table Size	Prefetch Distance	Cache Fill Level	L1D Misses	L2 Misses	LLC Misses	Software Prefetch Requests	Execution Time	Speedup
No Prefetching	7M	-	-	56418295	74923943	20887	0	459	
Software Prefetching	7M	2	L1	56623518	75266971	22587.6	1465	380	1.207894737
		2	L2	56709992.2	75332348.2	21947	720.6	369	1.243902439
		2	LLC	56636751.8	75332799.8	22809.6	0	369.6	1.241883117
		4	L1	56768578.6	75454067.2	25080.2	1465	378	1.214285714
		4	L2	56776294.8	75496935.	27263.	720.6	375	1.224

					4	6			
		4	LLC	56724122.4	75387526. 6	22303. 8	0	365	1.257534247
		8	L1	56710668.8	75389749. 6	22187. 6	1462.8	367.8	1.247960848
		8	L2	56765625.2	75281771. 4	22841. 8	718.6	364.8	1.258223684
		8	LLC	56745766.4	75387208	22823. 4	0	365.6	1.25547046
		16	L1	56661060.4	75304459. 4	24605. 8	1462.8	367	1.250681199
		16	L2	56706916.8	75378882. 8	28822. 2	718.4	371.4	1.235864297
		16	LLC	56647582.6	75246154. 6	24316	0	367	1.25
		32	L1	56747858.4	75387514. 6	22687. 4	1464	370	1.239200864
		32	L2	56743302	75504954. 6	26648. 6	719	376	1.220744681
		32	LLC	56818721	75559751. 4	27504. 8	0	393	1.166751398

Prefetching Mode	Embedding Table Size	Prefetch Distance	Cache Fill Level	L1D Misses	L2 Misses	LLC Misses	Software Prefetch Requests	Execution Time	Speedup
No Prefetching	8M	-	-	64461391	85660594	23172	0	442	
Software Prefetching	8M	2	L1	64817958	86192881	30481	1465	376	1.176783813
		2	L2	65018616	86276679	26164	721	381	1.160104987
		2	LLC	64883967	86270281	30412	0	379	1.16745906
		4	L1	64714405	85992810	33132	1465	373	1.185622318
		4	L2	64830374	86241607	25930	721	386	1.144484723
		4	LLC	64872400	86105050	24205	0	370	1.193304536
		8	L1	64804135	86075561	24601	1462	369	1.197183099
		8	L2	64760360	86036985	25125	719	377	1.173036093
		8	LLC	64839284	86166863	28705	0	374	1.181818182
		16	L1	64907543	86339460	28288	1463	373	1.185622318
		16	L2	64937075	86262450	24301	720	377	1.173659055
		16	LLC	64778080	86047139	25644	0	383	1.153444676
		32	L1	64797652	86109357	24276	1463	371	1.190732759
		32	L2	64923732	86251764	25361	719	367	1.203048449
		32	LLC	64711571	86017162	25703	0	372	1.189451023

Approach 2:

Running the prefetch loop for full embedding dimension.

Software_Prefetch Off:

Embedding Size	Cycles	Instructions	L1_Misses	L2_Misses	LLC_Misses	Execution Time
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1000000	3697940918	12105765151	8086529.6	10668198	11095.6	133.8
2000000	7401673711	24209403551	16130490.6	21265047	13106.4	137.2
3000000	11102835826	36316175115	24169643.2	31944808.8	14669.6	144.6

Task 2B: SIMD (Single Instructor Multiple Deadline)



Single *Instructor*, Multiple *Deadlines* (SIMD)

Utilize SIMD (Single *Instruction*, Multiple *Data*) instructions to parallelize the summation of elements within embedding rows, and analyze their impact on both instruction count and overall performance.

Deliverables

1. *Analyze the Impact of SIMD Instruction Width:*

Experiment with different SIMD instruction widths provided by Intel intrinsics (e.g., 64-bit, 128-bit, 256-bit), depending on your machine's capabilities, and observe how each affects performance.

2. *Analyze the Impact of Embedding Dimension:*

Try different embedding dimensions (i.e., number of elements per row) in combination with different SIMD widths.

3. *Analyze Instruction Count:*

Use the perf tool (or any suitable performance analysis tool) to measure the instruction count for different SIMD widths and embedding dimensions.

4. *Collect Execution Time and Compute Speedup:*

- Compute the speedup as follows:

$$Speedup = \frac{Execution\ Time\ without\ SIMD}{Execution\ Time\ with\ SIMD}$$

- Analyze how the speedup varies with different embedding table sizes and different software prefetching parameters.

Consider **Table 2.2** for reference. Identify the values of different metrics for different embedding dimensions and SIMD widths. Analyze them carefully and state your insights for the question below.

1. What trends do you observe in speedup for different combinations of embedding dimensions and SIMD widths?

SIMD width of 256 performs best based on speedup, until the point (2048 embedding dimension) where the speedup is drastically reduced compared to the 128 width SIMD register. Also, as the number of instructions are increasing, speedup is also increased for 256 bits, for 128 bits it is similar. In most cases, using higher dimensions does improve the speedup.

For which SIMD width do you achieve the maximum speedup?

256 bits

Table 2.2:

Metrics	SIMD width (bits)	embedding dimension (elements)	Instructions	Execution time	Speedup
No SIMD	-	128	43230594860	416	
SIMD	128		43223195325	351	1.185185185
	256		43221378493	261	1.5939
No SIMD	-	256	86474784240	708	
SIMD	128		86446170321	551	1.284936479
	256		86449188987	362	1.9558
No SIMD	-	512	172964705461	1255	
SIMD	128		172888167558	936	1.543705333
	256		172908306979	606	2.071
No SIMD	-	1024	345927273039	2406	
SIMD	128		345833282060	1760	1.367045455
	256		172884297210	590	4.078
No SIMD	-	1250	422166430273	2953	
SIMD	128		422187846136	2185	1.3515
	256		422153651382	1280	2.3070
No SIMD	-	1500	506622589915	3511	
SIMD	128		506652831662	2588	1.3566
	256		506611962390	1499	2.34223
No SIMD	-	2048	726988564251	733686	
SIMD	128		727581312490	519066	1.5135
	256		727525795477	569327	1.2899

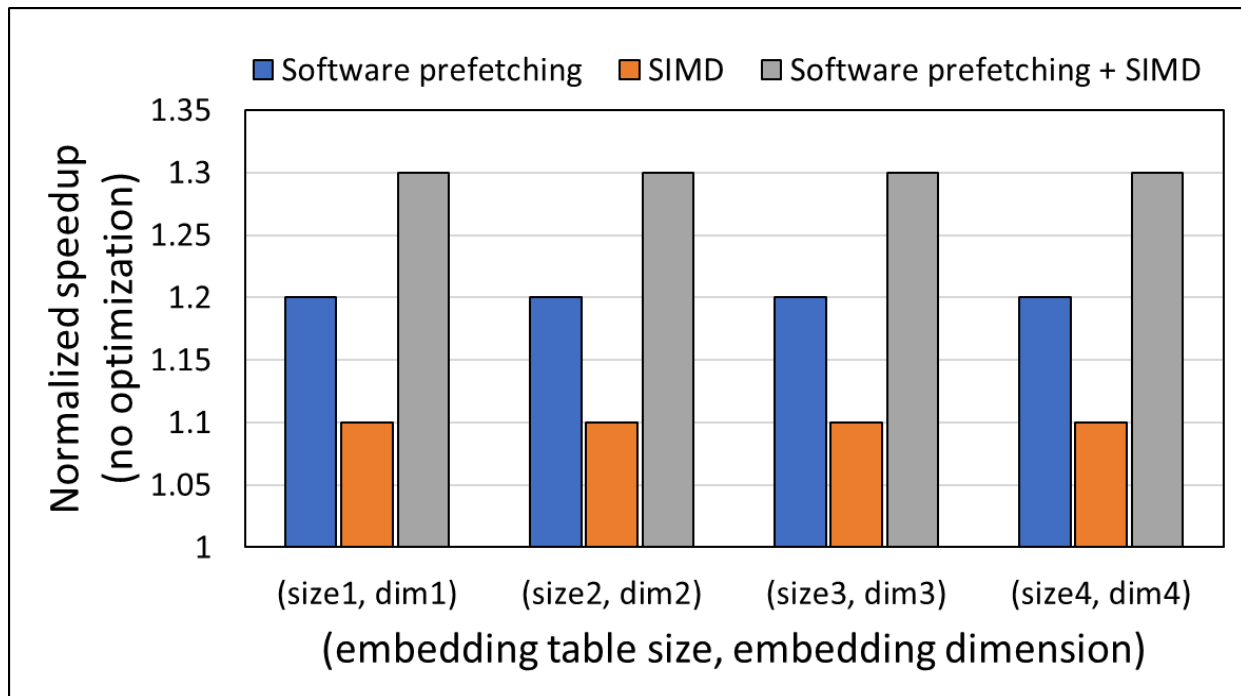
Task 2C: Software prefetching + SIMD

In this task, you will combine both software prefetching and SIMD techniques to optimize the embedding operation. You are required to repeat all the analyses you performed individually in Task 2A (Software Prefetching) and Task 2B (SIMD), but now with both optimizations applied together.

Deliverables

1. All the objectives from Task 2A and Task 2B.
2. Combine Table 2.1 and Table 2.2.

Figure 2.1:



Final performance summary for Embedding Operation:

Include a plot that compares the best performance achieved by each optimization technique –

- Software Prefetching
- SIMD
- Software Prefetching + SIMD

Plot this comparison against varying embedding table sizes and embedding dimension sizes. Refer to **Figure 2.1** for reference.

Table for 2C

Embedding Table Size	Dimensions	Registers (128b / 256b)	Prefetch Distance	Cache Fill Level	L1D Misses	L2 Misses	LLC Misses	Software Prefetch Requests	Execution Time	Speedup
1M	128	-	-	-	8159346	10861610	9738	0	418	
		256	2	L1	8120799	10788023	5835	1486	194	2.150980291
				L2	8182964	10970983	5670	742	195	2.147223506
				LLC	8137241	10824655	5760	0	189	2.207785348
			4	L1	8135383	10813568	5521	1485	194	2.150980291

				L2	8178704	10958993	5901	742	194	2.1583105 28
				LLC	8122476	10799735	5853	0	189	2.21164021 2
			8	L1	8150920	10819719	5578	1484	187	2.2313564 3
				L2	8130911	10812905	7602	740	213	1.9654864 34
				LLC	8197505	10874617	10257	0	211	1.9841458 2
			16	L1	8119313	10787901	8113	1483	214	1.9502636 12
				L2	8171360	10892800	7649	740	215	1.9471747 33
				LLC	8149699	10875480	7487	0	219	1.9058040 4
			32	L1	8119313	10787901	8113	1483	214	1.9502636 12
				L2	8124016	10793018	8322	739	226	1.8522621 53
				LLC	8138991	10814476	10052	0	215	1.9412065 2
			64	L1	8141333	10812278	11850	1479	229	1.8279616 92
				L2	8166949	10886281	8358	737	224	1.8633263 5

				LLC	8164626	10891549	8413	0	222	1.8828828 83
		128	2	L1	8147972	10823489	7590	1489	255	1.64133977 3
				L2	8156222	10888230	6960	743	256	1.6328125
				LLC	8138400	10821287	8269	0	261	1.60153256 7
			4	L1	8185024	10961320	7984	1489	271	1.54431595 7
				L2	8157674	10826659	6349	743	261	1.60356005 7
				LLC	8126868	10796968	7306	0	258	1.62015503 9
			8	L1	8139445	10820395	7448	1482	254	1.64566929 1
				L2	8157270	10838254	9178	739	255	1.63921568 6
				LLC	8190473	10977752	5669	0	270	1.54625827 7
			16	L1	8166515	10889223	6020	1486	275	1.5218261 91
				L2	8185965	10965490	5732	742	274	1.5255474 45

				LLC	8119508	10791015	5900	0	283	1.477031802
			32	L1	8170561	10888859	5407	1485	269	1.551999406
				L2	8176242	10924821	5583	739	268	1.557783327
				LLC	8132696	10807597	5310	0	271	1.542435424
			64	L1	8178202	10958246	5635	1481	275	1.521826191
				L2	8150798	10862767	5083	735	271	1.542435424
				LLC	8154265	10827330	5334	0	265	1.575396676

Embedding Table Size	Dimensions	Registers (128b / 256b)	Prefetch Distance	Cache Fill Level	L1D Misses	L2 Misses	LLC Misses	Software Prefetch Requests	Execution Time	Speedup
1M	512	-	-	-	73207162	82589564	265347	0	1330	

		256	2	L1	64706 109	858 821 31	311 45	1489	946	1.405919662
				L2	64867 587	860 554 69	280 67	743	926	1.436285097
				LLC	64771 908	860 260 31	358 04	0	942	1.411395159
			4	L1	64990 175	865 536 46	325 53	1487	951	1.398527865
				L2	64802 261	859 903 31	312 79	744	922	1.442000152
				LLC	64876 367	862 936 57	302 93	0	939	1.416400426
			8	L1	64689 883	858 571 11	249 69	1485	1010	1.317262076
				L2	64890 923	860 814 91	295 02	741	951	1.39804274
				LLC	64761 776	859 814 80	238 28	0	962	1.383010804

			16	L1	65032 189	866 712 91	294 51	1485	947	1.403945827
				L2	65078 024	860 774 55	319 04	741	941	1.412894522
				LLC	65084 354	867 168 82	312 42	0	943	1.410392365
			32	L1	64681 428	858 801 42	337 14	1487	963	1.380627615
				L2	64693 652	859 507 27	346 14	742	950	1.399513853
				LLC	64816 866	860 524 93	248 96	0	942	1.411889597
			64	L1	64900 141	860 766 23	255 78	1483	964	1.379195918
				L2	64917 031	865 051 22	246 84	741	969	1.37301661
				LLC	64879 467	860 959 99	268 52	0	973	1.366443036

		128	2	L1	32399 436	429 954 79	142 37	1492	821	1.619324754
				L2	32457 097	430 741 76	142 75	746	814	1.633244508
				LLC	32397 067	430 056 33	160 89	0	827	1.607581014
			4	L1	32593 854	436 511 48	179 74	1490	836	1.591537329
				L2	32540 476	433 664 12	142 13	744	819	1.624586219
				LLC	32524 457	434 043 11	160 33	0	829	1.604981476
			8	L1	32458 546	430 687 53	146 10	1486	810	1.641306628
				L2	32363 511	429 676 38	137 06	743	829	1.603704195
				LLC	32495 025	431 030 53	167 43	0	811	1.639950678

			16	L1	32465 964	430 431 59	161 60	1486	828	1.606920633
				L2	32809 833	438 879 21	142 68	741	843	1.578316541
				LLC	327113 07	432 863 77	155 06	0	842	1.579572447
			32	L1	32473 935	431 256 76	151 28	1484	814	1.634569297
				L2	32379 655	429 969 41	137 62	742	819	1.623277556
				LLC	32445 786	430 527 85	152 02	0	820	1.621298746
			64	L1	32356 271	429 630 14	132 87	1483	831	1.59984603
				L2	32420 988	865 051 22	430 414 29	14036	739	1.798926055
				LLC	32543 031	433 492 12	148 46	0	830	1.603046995

Embedding Table Size	Dimensions	Registers (128b / 256b)	Prefetch Distance	Cache Fill Level	L1D Misses	L2 Misses	LLC Misses	Software Prefetch Requests	Execution Time	Speedup
1M	1024	-	-	-	144 140 842	491 247 759	489 086	0	2475	
		256	2	L1	324 546 91	431 972 80	516 09	1487	535	4.626168224
				L2	324 423 59	430 963 62	214 40	743	531	4.661016949
				LLC	324 306 51	430 491 87	134 19	0	525	4.714285714
			4	L1	323 835 61	429 940 48	139 65	1488	528	4.684572142
				L2	326 560 08	437 077 92	129 58	743	527	4.693455711
				LLC	326 089 49	434 047 35	140 19	0	567	4.365079365
			8	L1	324 132 08	430 488 41	129 41	1485	532	4.652255639
				L2	324 892 61	433 236 66	134 50	743	521	4.750479846
				LLC	323 767 70	429 798 57	134 21	0	541	4.577653652
			16	L1	324 709 21	430 560 47	127 47	1485	537	4.606107978

				L2	323 759 89	429 831 36	136 03	738	558	4.438108559
				LLC	324 389 81	430 538 44	137 98	0	534	4.631969008
			32	L1	326 078 08	434 257 55	132 67	1490	534	4.637697453
				L2	326 304 73	436 472 48	130 48	739	541	4.574861368
				LLC	324 752 19	433 010 95	132 74	0	533	4.640654004
			64	L1	326 089 69	431 588 66	136 37	1482	532	4.655143228
				L2	324 399 56	430 464 49	139 88	739	538	4.600371747
				LLC	325 384 25	434 064 54	161 69	0	527	4.693455711
		128	2	L1	651 962 95	871 999 11	250 24	1488	1557	1.589595376
				L2	653 845 57	875 533 61	240 01	744	1571	1.575760663
				LLC	649 586 61	862 570 41	248 07	0	1532	1.615883317

			4	L1	648 009 58	860 824 93	245 40	1489	1531	1.616938988
				L2	652 655 99	867 886 33	233 47	742	1559	1.587220152
				LLC	650 933 41	867 395 96	235 46	0	1547	1.599529512
			8	L1	650 446 36	866 652 80	233 63	1488	1566	1.580792887
				L2	650 337 37	868 732 73	236 33	742	1539	1.607842373
				LLC	647 933 86	859 543 16	252 82	0	1557	1.589258539
			16	L1	648 028 71	859 874 58	246 18	1486	1533	1.614829024
				L2	649 560 02	865 381 20	285 37	743	1568	1.578776145
				LLC	647 329 48	859 374 87	260 47	0	1568	1.578111749
			32	L1	652 045 56	867 133 96	268 19	1486	1553	1.594028351
				L2	649 734 95	865 534 38	265 28	742	1563	1.583159026
				LLC	649 339 13	861 115 50	304 95	0	1543	1.60436127
			64	L1	647 248 19	859 053 05	296 83	1485	1567	1.579451181
				L2	651 461 33	867 684 27	326 37	740	1569	1.577106154

				LLC	648 261 32	860 817 07	290 83	0	1563	1.58382768
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Embedding Table Size	Dimens ions	Register s (128b / 256b)	Prefe tch Dista nce	Cach e Fill Level	L1D Misses	L2 Misses	LLC Misses	Software Prefetch Request s	Executio n Time	Speedup
1M	1500	-	-	-	213915142	2390873 58	665106	0	3582	
		256	2	L1	94914630	1260377 27	50071	1490	1386	2.585031068
				L2	95043067	1267175 50	43879	759	1411	2.539218953
				LLC	95077483	1262464 53	36126	0	1380	2.595652174
			4	L1	95574877	1276824 48	44696	1489	1430	2.505473291
				L2	95389792	1269931 46	41775	756	1401	2.556143093
				LLC	94774130	1257988 40	52855	0	1406	2.547652916
			8	L1	95042662	1261837 09	53181	1489	1399	2.560400286
				L2	94867298	1258010 17	43313	759	1425	2.513102229
				LLC	95170125	1261418 98	36013	0	1429	2.507227001
			16	L1	95532280	1278474 11	48387	1489	1436	2.495002333
				L2	94961231	1259792 09	39543	759	1412	2.536827195
				LLC	94851773	1259445 91	54431	0	1428	2.507823822
			32	L1	95013142	1260882 33	53208	1488	1419	2.524312896
				L2	94941857	1260068 63	37204	748	1402	2.554921541

				LLC	95261630	1269242 96	38740	0	1411	2.539218953
			64	L1	94866656	1258947 31	56831	1487	1437	2.493265677
				L2	95088299	1260609 39	51084	756	1428	2.508983168
				LLC	95268509	1269592 49	50002	0	1465	2.445051195
		-	-	-						
		128	2	L1	95160672	1267597 93	40335	1485	2301	1.556714472
				L2	95536372	1278234 72	43382	742	2305	1.55423553
				LLC	95269650	1267920 70	39156	0	2297	1.559425337
			4	L1	94810342	1258734 17	48613	1485	2310	1.550870904
				L2	95074251	1260738 83	39599	741	2293	1.562370511
				LLC	94908374	1259384 26	37932	0	2304	1.554464855
			8	L1	95019899	1260647 70	41367	1488	2299	1.558068726
				L2	95452145	1277153 48	43766	740	2313	1.548859111
				LLC	95058806	1260017 49	42267	0	2294	1.56146469
			16	L1	95131331	1268070 45	46630	1483	2319	1.544851143
				L2	95039688	1260748 81	44753	738	2309	1.551542663
				LLC	95075096	1260115 68	50990	0	2296	1.559880331
			32	L1	94982857	1259503 56	53684	1483	2343	1.529024575

				L2	95458149	1278503 92	36370	738	2328	1.538659794
				LLC	95316706	1269020 46	37623	0	2317	1.545744456
			64	L1	95732691	1276521 03	51859	1481	2330	1.537339056
				L2	94983799	1260621 75	48242	738	2316	1.546632124
				LLC	95926002	1284804 78	46491	0	2353	1.522311942

Embedding Table Size	Dimensio ns	Registers (128b / 256b)	Prefetc h Distanc e	Cache Fill Level	L1D Misses	L2 Misses	LLC Misses	Softw are Prefet ch Requ ests	Execu tion Time	Speedup
2M	128	-	-	-	36549611	41683282	15518 5	0	430	
		256	2	L1	16167320	21503077	13527	1486	223	1.928251121
				L2	16183079	21508293	16585	742	201	2.139303483
				LLC	16295415	21617056	11554	0	197	2.182741117
			4	L1	16259849	21581821	10549	1485	204	2.107843137
				L2	16175840	21479205	5901	10731	242	1.776859504
				LLC	16180987	21473210	8785	0	201	2.139303483
			8	L1	16271303	21627551	11872	1481	203	2.118226601
				L2	16257252	21574729	9295	739	208	2.067307692
				LLC	16327413	21935601	11055	0	213	2.018779343
			16	L1	16177605	21498012	11824	1484	214	2.006252041
				L2	16263992	21611698	10771	740	214	2.009345794
				LLC	16321884	21730474	8891	0	216	1.990740741
			32	L1	16165960	21469096	8636	1482	217	1.98156682
				L2	16158055	21453987	9884	740	218	1.972477064
				LLC	16268729	21602759	10815	0	218	1.972477064
			64	L1	16220042	21477465	8696	1484	240	1.791666667
				L2	16311386	21646283	10619	740	217	1.98156682
				LLC	16180668	21483748	10212	0	222	1.936936937

		128	2	L1	16174880	21475256	12241	1487	277	1.55234657
				L2	16287564	21616738	13597	1487	276	1.557971014
				LLC	16155436	21458456	11077	0	257	1.673151751
			4	L1	16170054	21490204	11371	1485	267	1.610486891
				L2	16259814	21582301	11923	742	259	1.66023166
				LLC	16246358	21572961	10196	0	279	1.541218638
			8	L1	16336577	21948288	10075	1483	281	1.53024911
				L2	16256140	21590170	8902	739	264	1.628787879
				LLC	16162205	21475576	8674	0	286	1.503496503
			16	L1	16259420	21569682	10052	1484	272	1.580882353
				L2	16173623	21483466	9953	738	297	1.447811448
				LLC	16323750	21943480	8822	0	275	1.563636364
			32	L1	16242192	21565786	10449	1483	281	1.53024911
				L2	16302780	21880543	11020	739	303	1.419141914
				LLC	16266499	21593264	9354	0	282	1.524822695
			64	L1	16264983	21581481	14676	1479	274	1.569343066
				L2	16173226	21475943	9778	738	270	1.592592593
				LLC	16238974	21545840	7772	0	269	1.598513011

Embedding Table Size	Dimensions	Registers (128b / 256b)	Prefetch Distance	Cache Fill Level	L1D Misses	L2 Misses	LLC Misses	Software Prefetch Requests	Execution Time	Speedup
2M	512	-	-	-	145858250	163779594	492228	0	1307	
		256	2	L1	65151298	86204182	25352	1489	547	2.389396709
				L2	64684821	85671970	22388	745	554	2.359205776
				LLC	64978244	86146750	23369	0	519	2.518304432
			4	L1	64530919	85661308	25095	1490	543	2.406998158
				L2	65044195	86272678	21828	748	532	2.456766917
				LLC	65213718	87662786	32165	0	525	2.48952381
			8	L1	64856507	86027127	24256	1483	527	2.480075901
				L2	64531661	85680180	25924	738	529	2.470699433
				LLC	64613524	85947260	29599	0	523	2.499043977
			16	L1	65145389	86334871	25621	1486	558	2.342293907
				L2	64845553	86171672	34260	737	547	2.389396709

				LLC	65181010	87575360	25584	0	547	2.389396709
			32	L1	65132532	86349847	23012	1486	533	2.452157598
				L2	64589772	85728963	25127	741	560	2.333928571
				LLC	64857879	86073885	29967	0	541	2.415896488
			64	L1	64564826	85752663	30104	1482	527	2.480075901
				L2	64955055	86190873	25480	739	531	2.461393597
				LLC	64606194	85771184	29011	0	532	2.456766917
		128	2	L1	64915412	86108842	23235	1490	824	1.586165049
				L2	64931942	86162853	25583	745	825	1.584242424
				LLC	65212495	87588621	24781	0	818	1.597799511
			4	L1	64637251	86014994	38341	1487	866	1.509237875
							14270			
				L2	65073150	86437558	0	743	826	1.582324455
				LLC	64838920	86035953	22390	0	817	1.599755202
			8	L1	64586883	85702118	38607	1485	849	1.539458186
				L2	64545993	85708799	22815	744	838	1.559665871
				LLC	64920971	86077165	21770	0	829	1.576598311
			16	L1	64963373	86162170	23248	1485	830	1.574698795
				L2	64921553	86130316	22966	742	833	1.569027611
				LLC	64623801	85799340	23798	0	845	1.546745562
			32	L1	64559126	85647032	24045	1484	854	1.530444965
				L2	64996415	86110168	21528	743	819	1.595848596
				LLC	64519787	85643621	25407	0	875	1.493714286
			64	L1	64853330	86070071	23201	1481	821	1.591961023
				L2	64532350	85666612	23406	736	857	1.525087515
				LLC	64982962	86353701	81986	0	829	1.576598311

Embedding Table Size	Dimensions	Registers (128b / 256b)	Prefetch Distance	Cache Fill Level	L1D Misses	L2 Misses	LLC Misses	Software Prefetch Requests	Execution Time	Speedup
2M	1024	-	-	-	143758910	1335262524	4174294	347	435234	

		256	2	L1	173555171	255123611	2806004	987	967620	0.4497984746
				L2	165616233	223892256	290539	53	7804179	0.0557693512
				LLC	166684621	219822834	198887	0	1816444	7
			4	L1	163775912	218087972	170860	981	490666	0.2396077171
				L2	167069319	216699504	156555	465	605259	0.8870270204
				LLC	179793147	261084410	3347824	0	563675	0.7190872007
			8	L1	170234208	221102459	149193	1177	452648	0.7721364261
				L2	165331914	211925059	141703	446	491646	0.961528605
				LLC	171989043	255038747	2411783	0	423397	0.8852589058
			16	L1	160491885	212507370	143429	1190	448507	1.027957213
				L2	163704296	213836261	138005	434	480117	0.970406259
				LLC	162577265	217604377	168612	0	539731	0.9065165366
			32	L1	161759605	208889095	154097	1176	453338	0.8063905909
				L2	159566272	209234305	139559	399	572293	0.960065117
				LLC	165657277	211813942	141671	0	488231	0.7605090399
										0.891450973
		128	2	L1	178110040	260754887	3557673	904	598588	0.7271011113
				L2	164255697	214617112	154107	244	504845	0.8621141142
				LLC	167919925	256750374	3421963	0	427229	1.018737024
			4	L1	160828497	210444135	140765	1005	539656	0.806502661
				L2	166740836	210714687	145523	452	480428	0.9059297127
				LLC	168372194	253448643	3344069	0	873383	0.9059297127
			8	L1	186762183	262894294	2746307	1214	407342	0.4983312018
				L2	161882472	214797311	142482	81	496435	1.068473175
				LLC	176045812	262889191	4314653	0	454185	0.8767190065
			16	L1	161103082	215038511	141287	1040	407590	0.9582747118
				L2	162456603	210818282	142788	429	553206	1.067823057
				LLC	168833042	251690430	2624803	0	1002760	0.7867485168
			32	L1	165852975	215089623	160113	924	573010	0.4340360605
				L2	165126813	215048499	150151	458	487785	0.7595574248
				LLC	163028414	214283403	150913	0	441225	0.8922660598
			64	L1	169326206	254753030	3140022	975	487179	0.9864218936
				L2	178568047	254215895	2852600	427	537692	0.893375946
				LLC	164376423	215007414	141471	0	1413002	0.8094485319
										0.3080207954

Extra Optimizations:

1. Using the array to speedup up the sum operations across the embedding rows along with SIMD.

```
// Divide in 8 chunks
int iter = embedding_dim / 8;
__m256 store[128];

// Initialize the array
for (int v = 0; v < iter; v++) {
```

```

    store[v] = _mm256_setzero_ps();
}

for (int j = start_idx; j < end_idx; ++j) {
    const float* data_ptr = &embedding_table[input[j] * embedding_dim];

    int offset = 0;

    for(int i=0; i < iter; i++) {

        // Load 8 elements at a time
        __m256 curr = _mm256_loadu_ps(&data_ptr[i * 8]);

        // Perform addition
        store[i] = _mm256_add_ps(store[i], curr);
    }

    // Summing remaining parts
    for (int k = iter * 8; k < embedding_dim; k++) {
        bag_embedding[k] += data_ptr[k]; } }

```